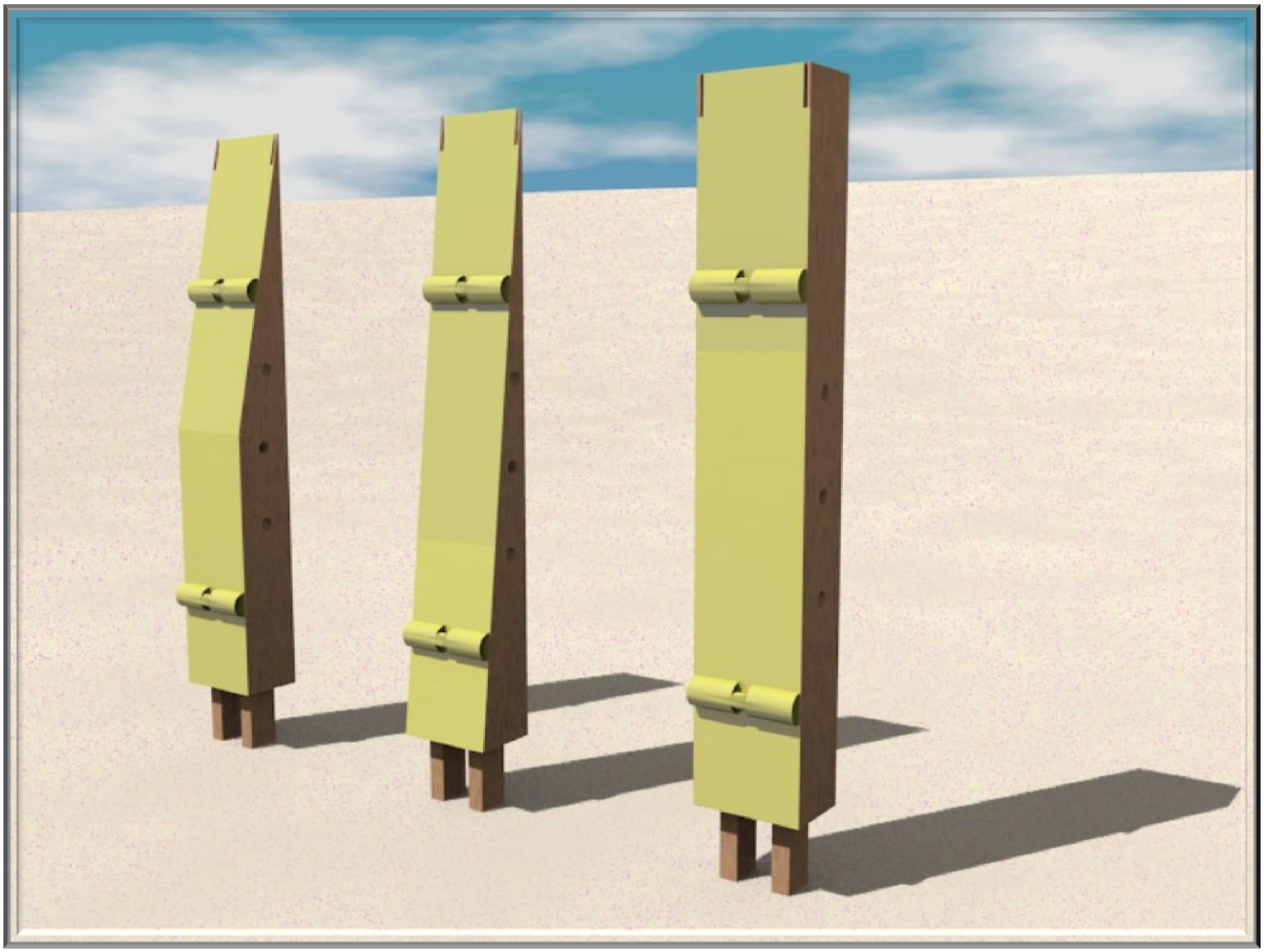
בס"ד

The Shape of the Mishkan Beams



Based on Talmud Bavli, Masechta Shabbos, page 98

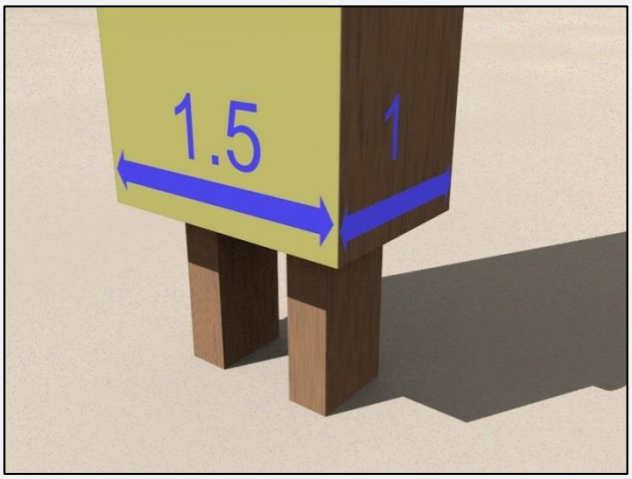
Presented by Bumy Goldson - whenevertobe@gmail.com

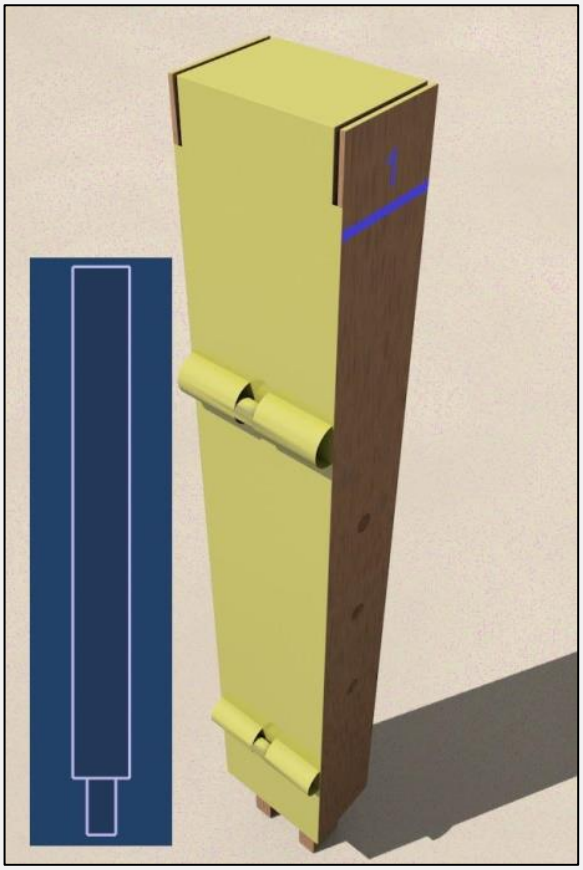
חיים אברהם גולדסאן

**All measurements are in amos (cubits), which is about a foot and a half or two (about 1/2 of a meter).*

Introduction

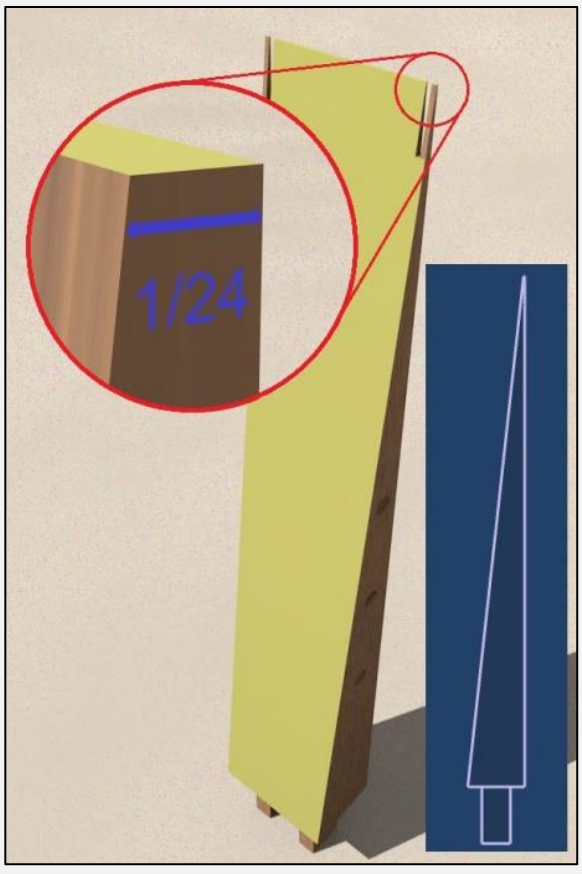
The bottoms of the beams in the Mishkan were 1½ amos wide[1] and 1 amah thick[2].





Rebbe Nechemia says that just like the beams were 1 amah thick at the bottom, so too they were 1 amah thick[3a] at the top.

Rebbe Yehuda says that the beams tapered off towards the top[3b] until they were only an etzba at the very top (1/24 of an amah).



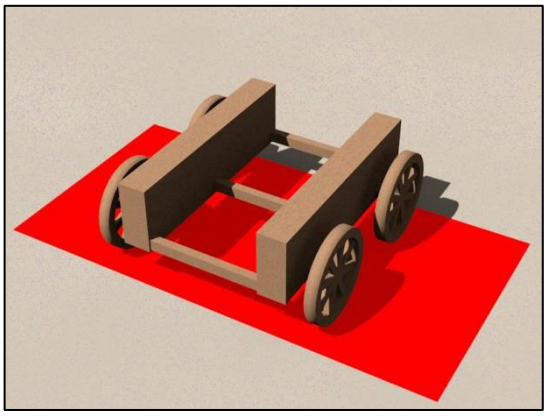
The Contradiction between Two Statements of Rav

Rav said the following statements:

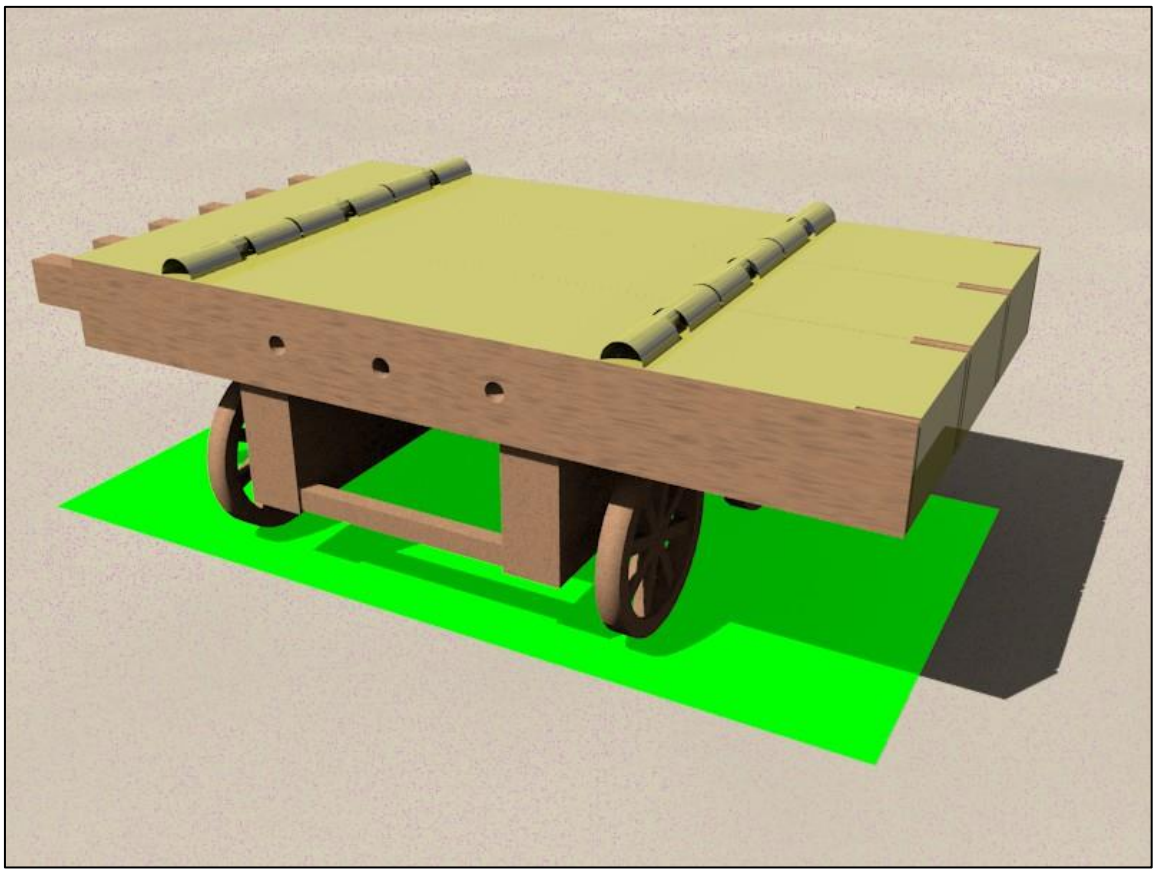
1. "Carrying within a roofed public domain is exempt from the laws of carrying on Shabbos."

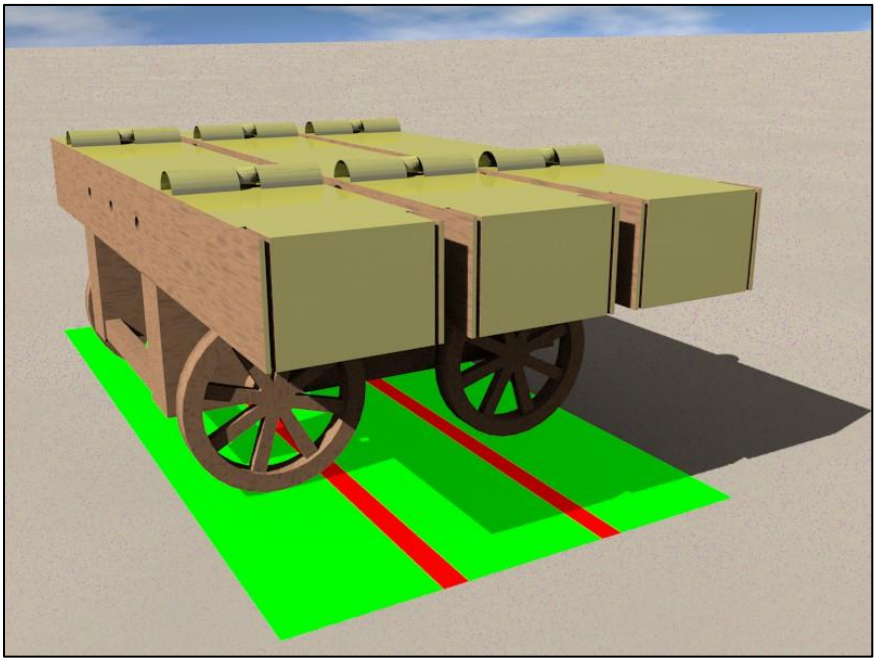


2. "Carrying along the sides of the wagons that carried the beams of the Mishkan was forbidden on Shabbos."



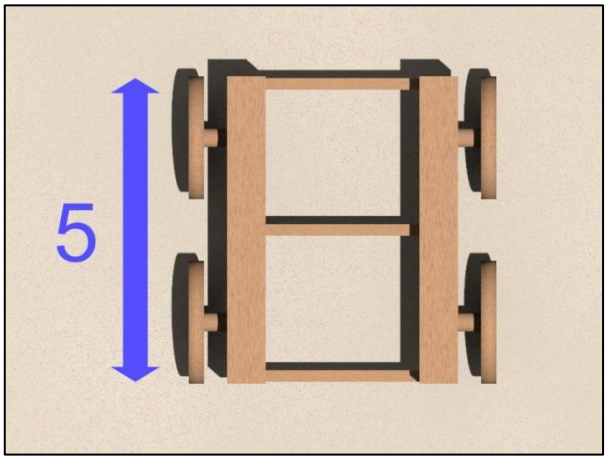
Considering how the beams jutted out over the sides of the wagons, creating a roof, **why should it be forbidden to carry there**?



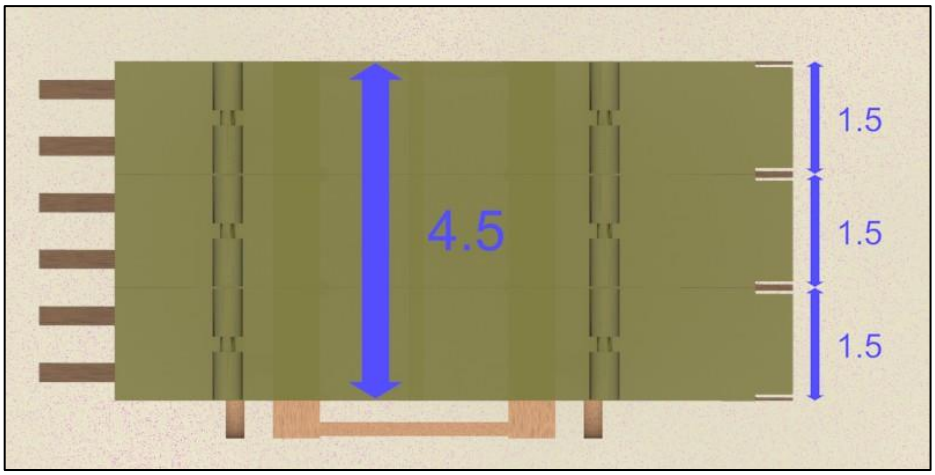


The Gemara answers that when Rav said that carrying alongside the wagons was forbidden, he meant in the areas underneath the spaces between the beams.

The Gemara then calculates exactly how much space there was between the beams.

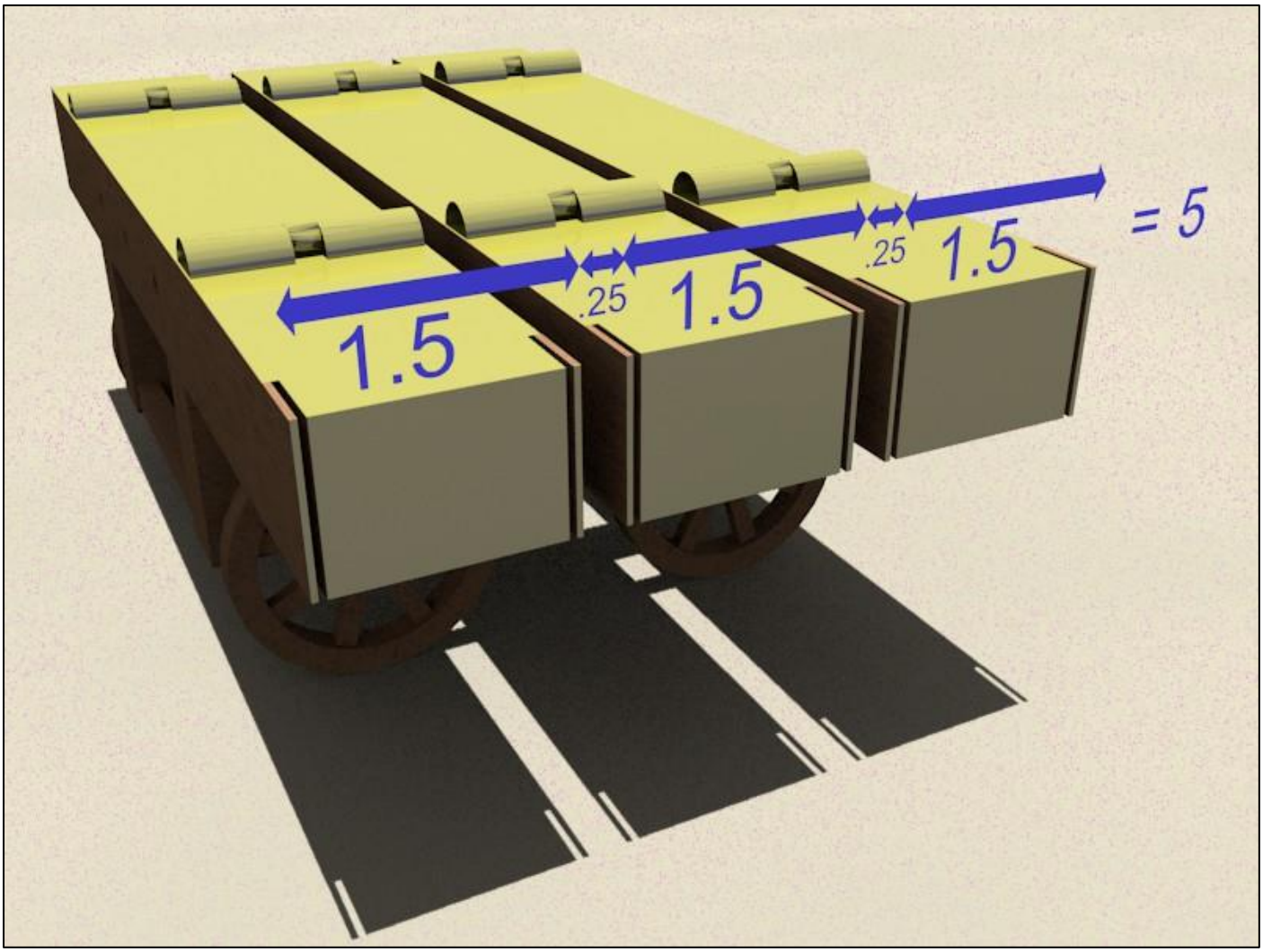


The length of the wagon was 5 amos long[4].

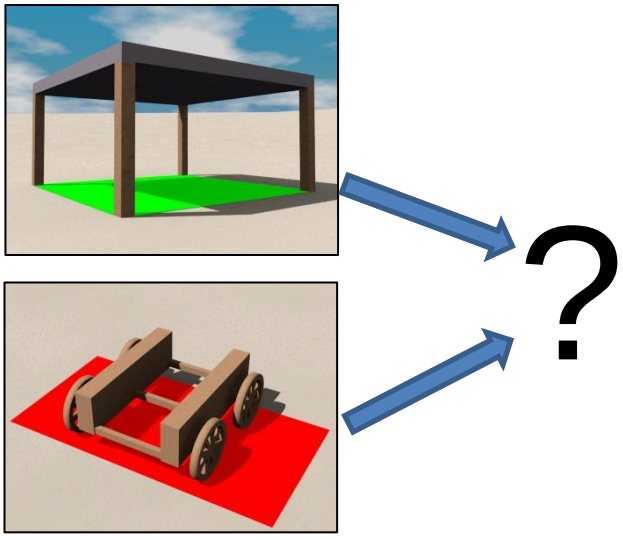


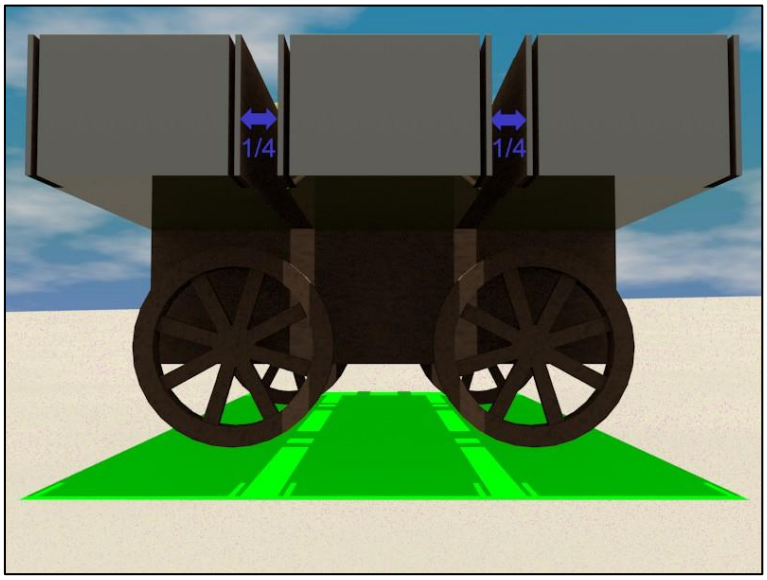
The beams were placed lengthwise across the width of the wagon. Each beam was 1 and a half amos wide. Only three beams could fit across the length of the wagon, leaving half of a amah uncovered.

Assuming that they spaced the beams out across the whole length of the wagon we would find that there was 1/4 of an amah between each beam.

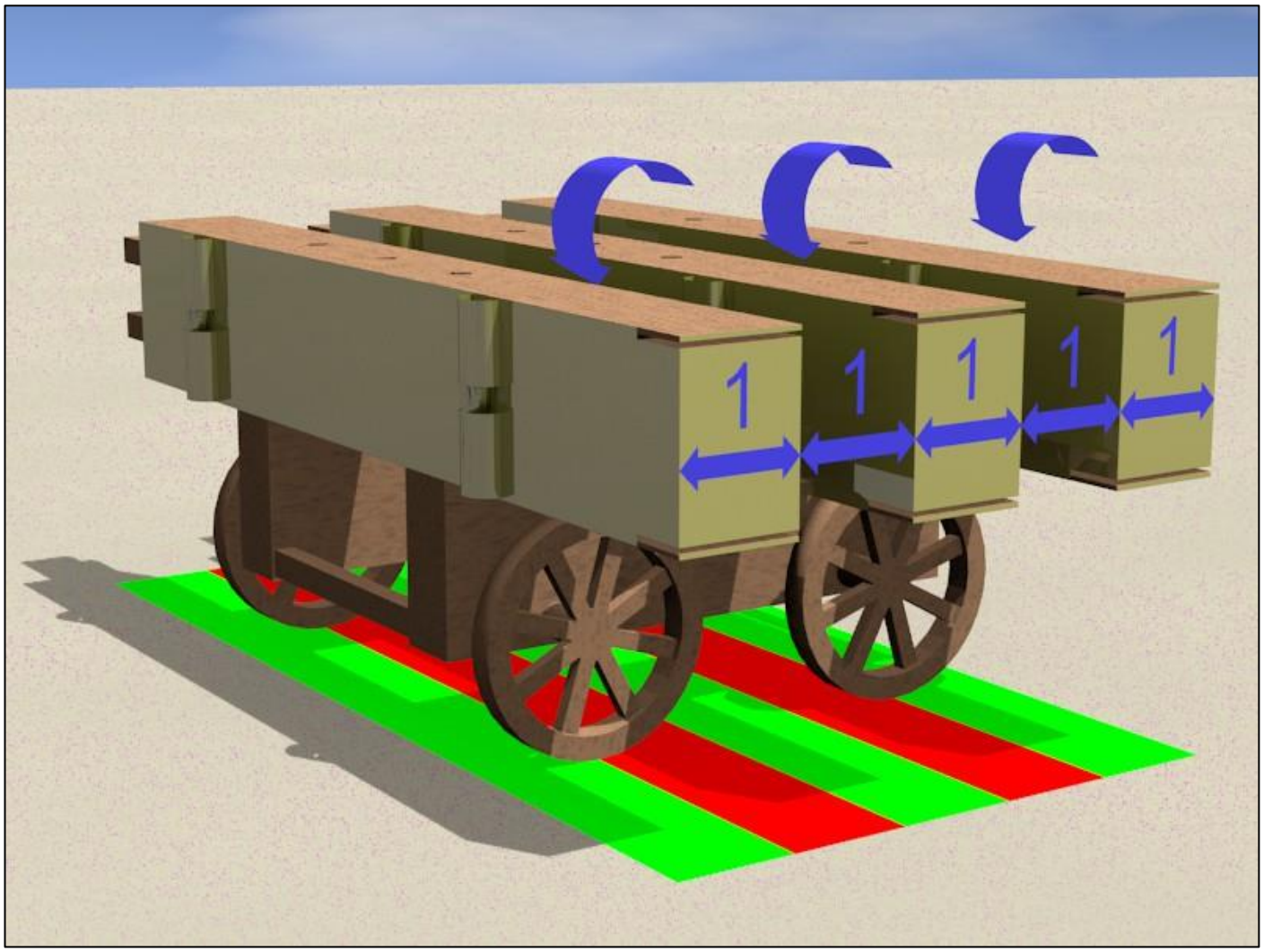


The problem is that there is a concept known as *"extension"* (*lavud,* לבוד), which determines that **a space less than 1/2 of an amah is considered as closed**[5]. This brings us back to our original problem. If the entire area along the sides of the wagons was considered "roofed" then how can Rav say that it was forbidden to carry there on Shabbos when he himself said that one is exempt from carrying within a roofed area?





The Gemara responds that the beams were not lying face down. Rather, they were lying on their sides, increasing the amount of space between each beam to 1 amah, negating the concept of *"extension"* and leaving that area uncovered.

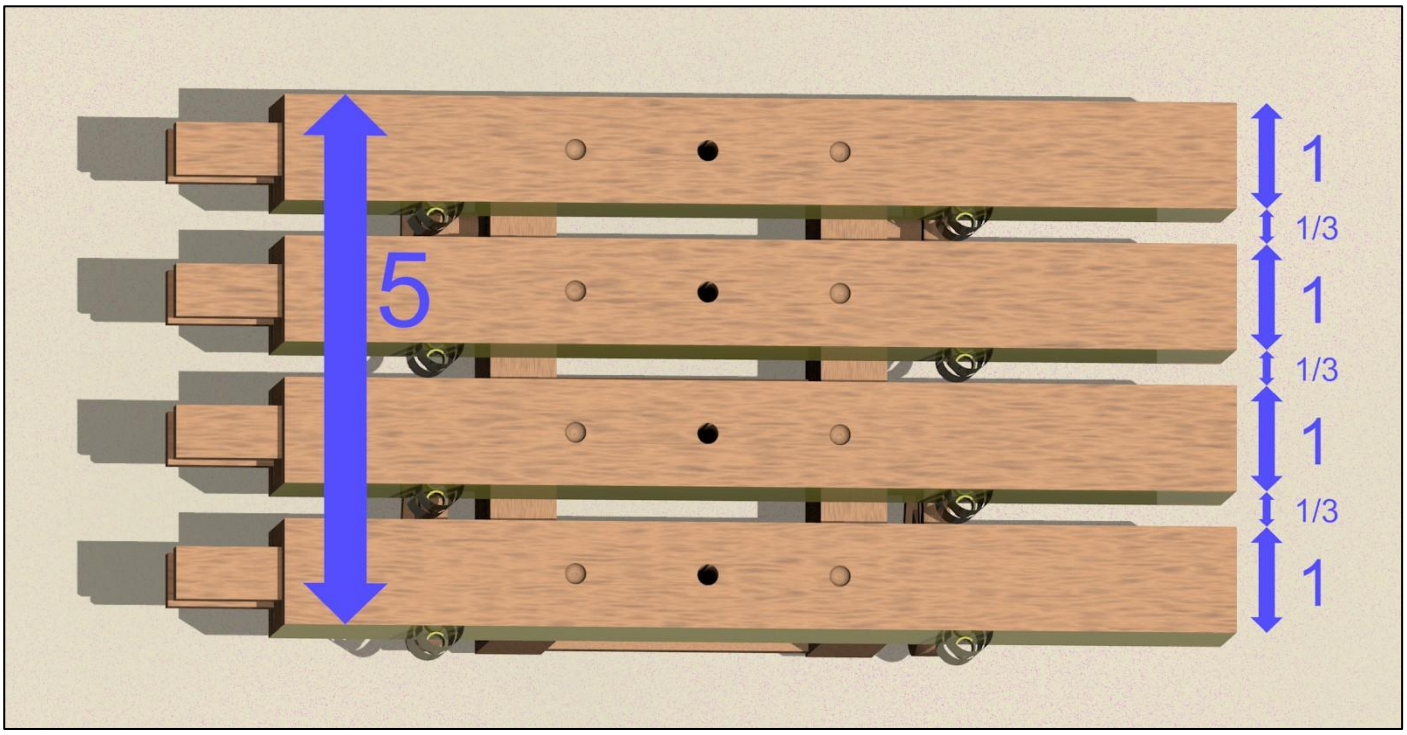


It was in this uncovered area that Rav was referring to when he said that it was forbidden to carry there.

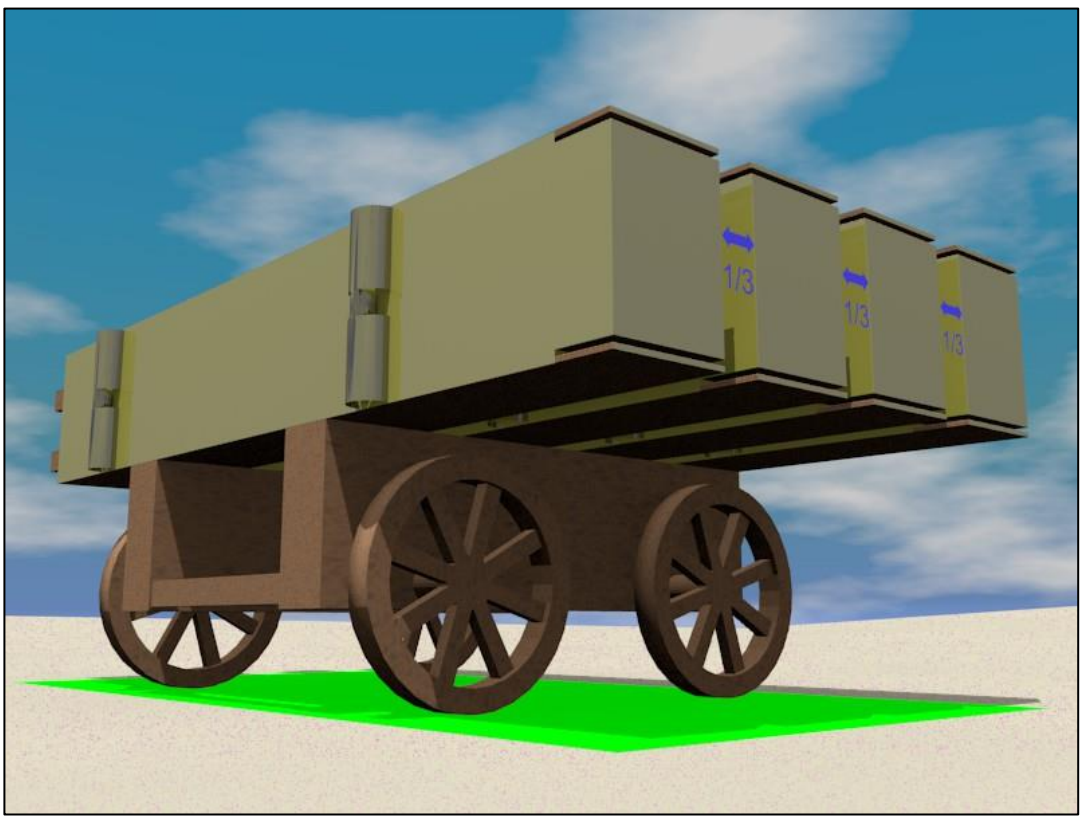
Still, this does not solve the problem. If they were laid on their sides then there would have been enough room for one more beam, leaving 1 amah of uncovered ground.

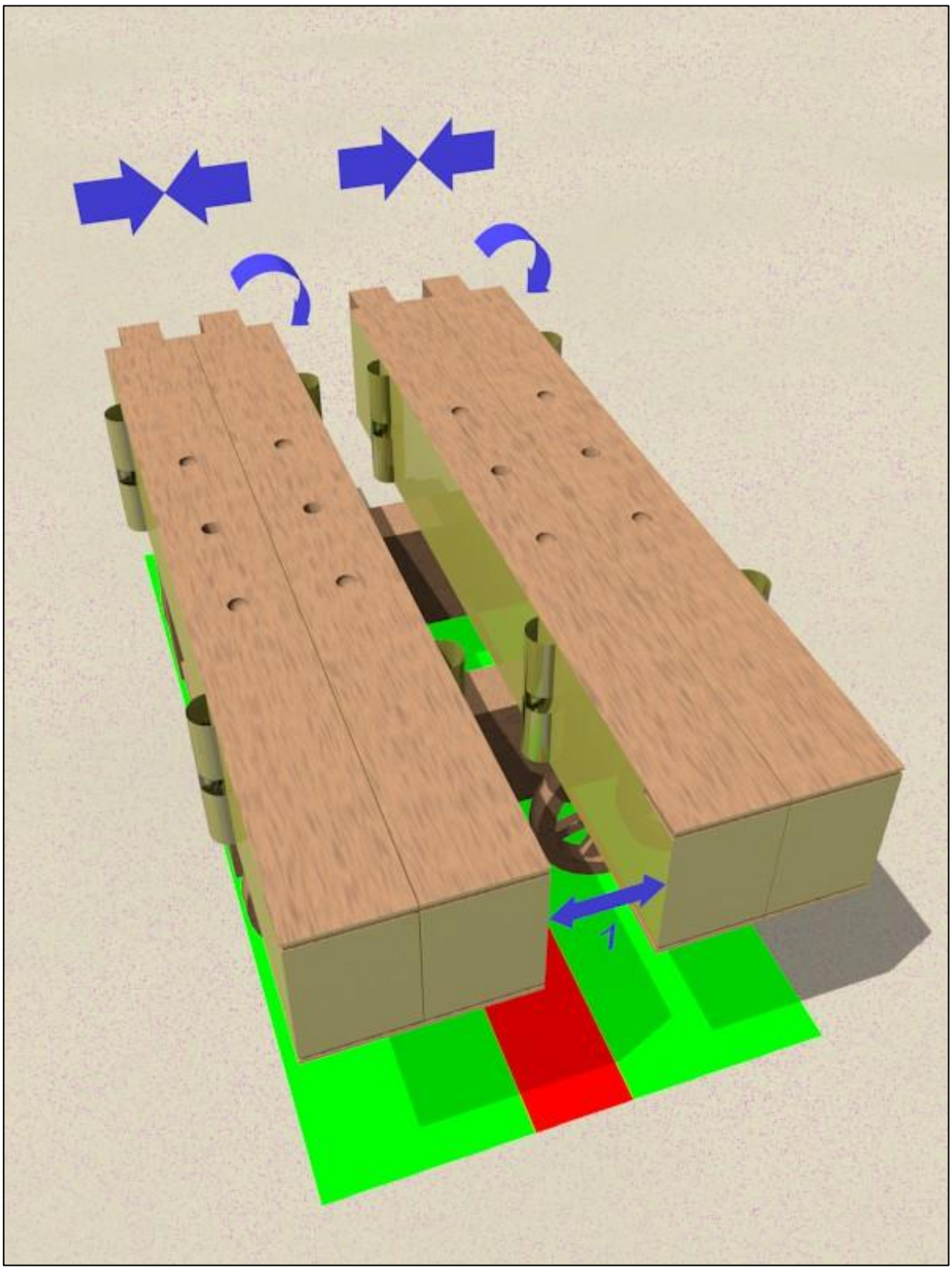


Spread out evenly, we would find that there would be 1/3 of an amah between each beam, thus reactivating the concept of *"extension"* (which operates on any space less than 1/2 of an amah).



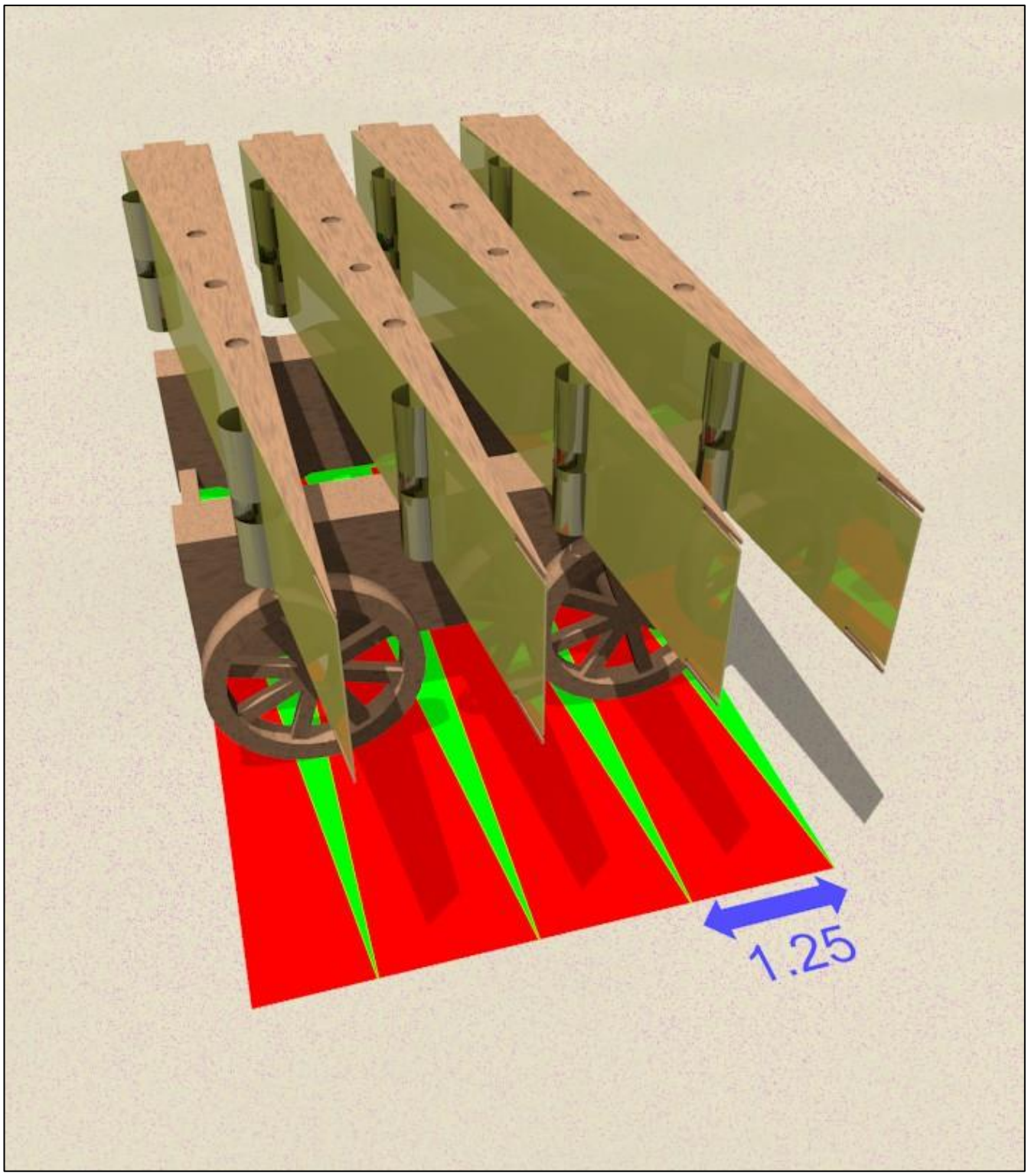
So, we are back to our original problem again. How can Rav say that it is forbidden to carry along the sides of the wagons if he also says that one is exempt from carrying underneath a roof?





The Gemara answers that the beams were grouped together at the ends of the wagon, back to back[6]. Now, there would be 1 amah of space between the wagons, which is too much to be considered closed.

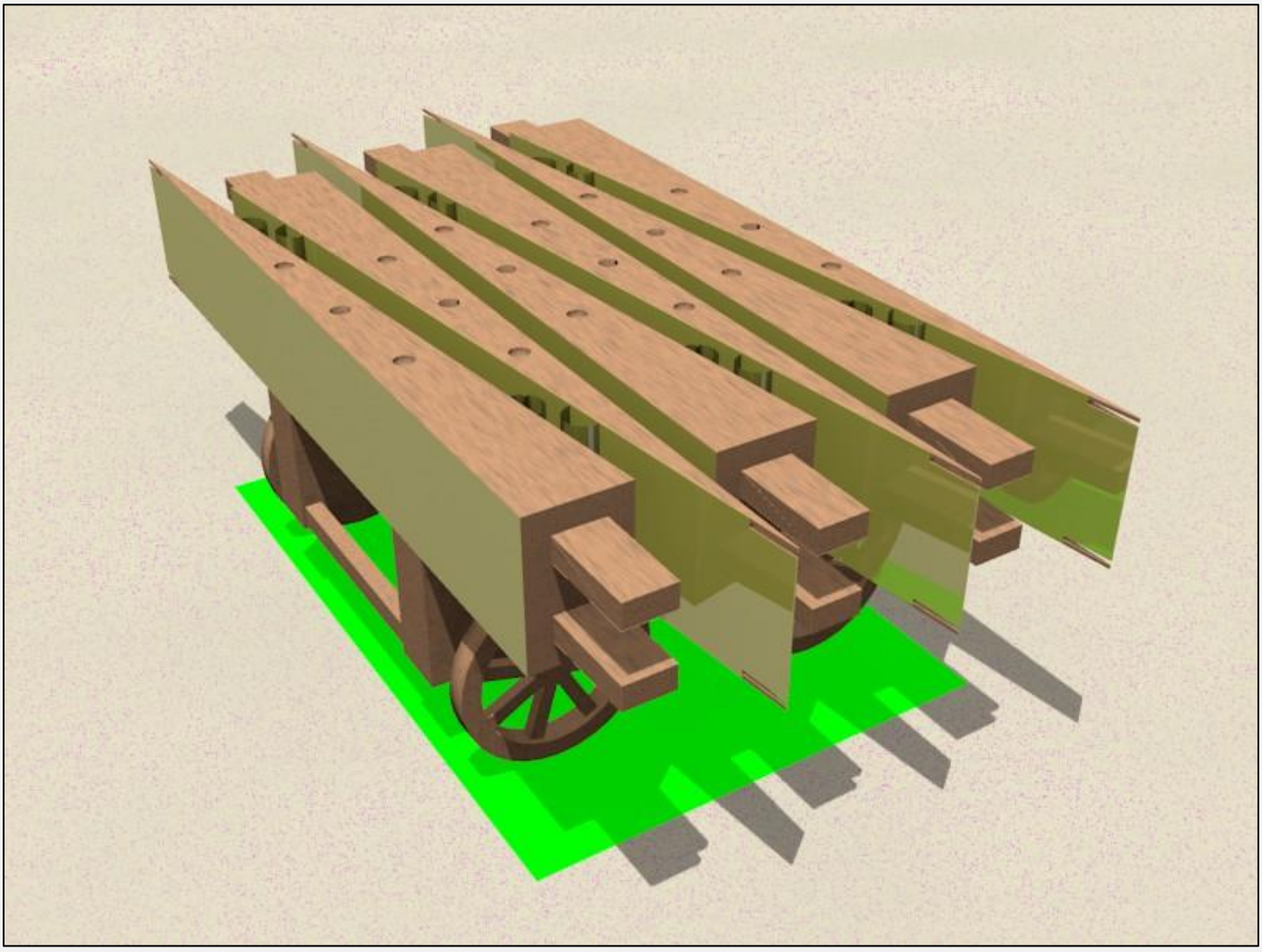
The Gemara points out that according to Rebbe Yehuda, who we brought in the introduction, we don't need to say that the beams were arranged in this way because according to him the beams tapered off which left enough space between the beams that can be considered open space.



Accordingly, even though Rav said that one is exempt from carrying underneath a roofed area, this does not contradict his other statement that at the sides of the wagon it is forbidden, since we can say that he was referring to the area beneath the open spaces between the thin ends of the beams.

Tosafos's Question and Answers

The Gemara said that according to Rebbe Yehuda, who says that the beams tapered off at the top, there would be no contradiction between the two statements of Rav. However, asks Tosafos[7], if the beams were indeed shaped that way then why wouldn't they have loaded the beams onto the wagons in an alternating fashion, thick end to thin end?

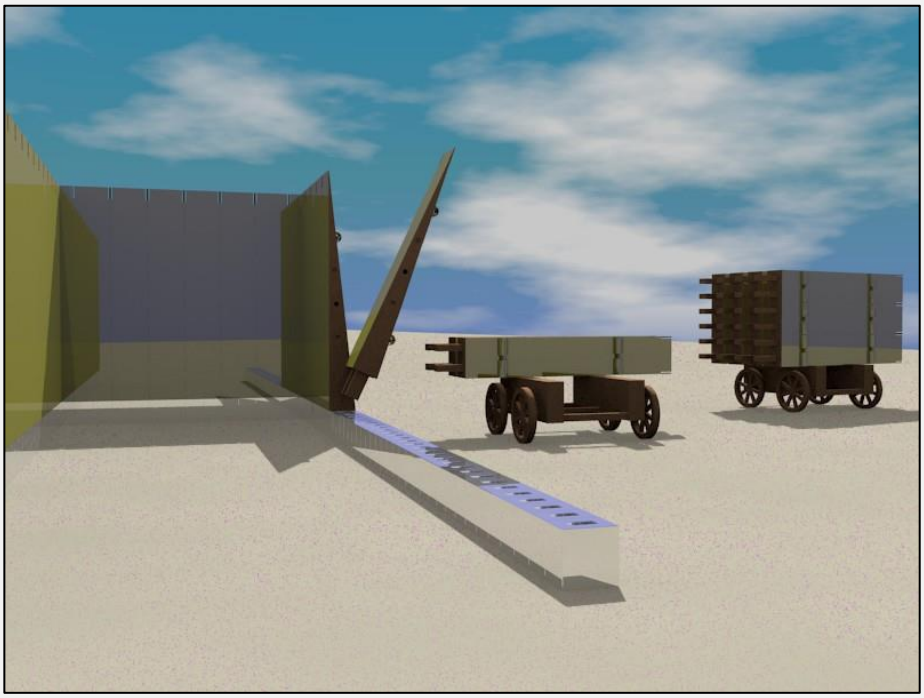


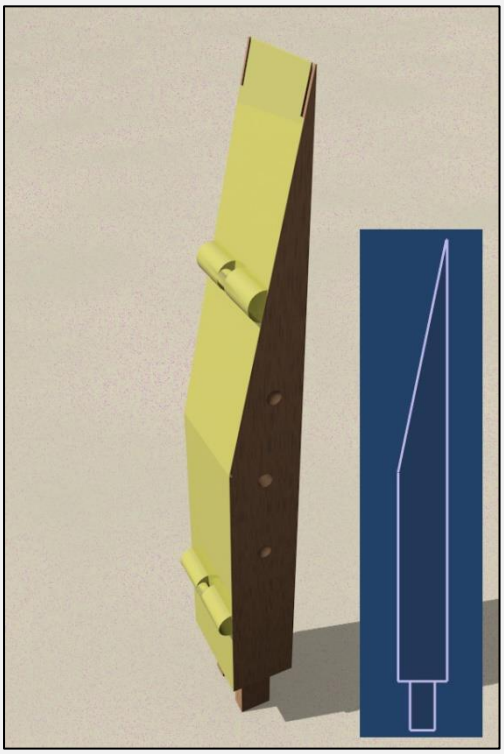
You could fit at least six beams on a single row that way and then we would return to our original problem, yet again! Once again, the beams were close enough to be considered a roofed area[8]. Why, then, does Rav say that during Shabbos it is forbidden to carry along the sides of the wagons if he himself maintains that a roofed area is an exempt area?

Tosafos suggests three answers:

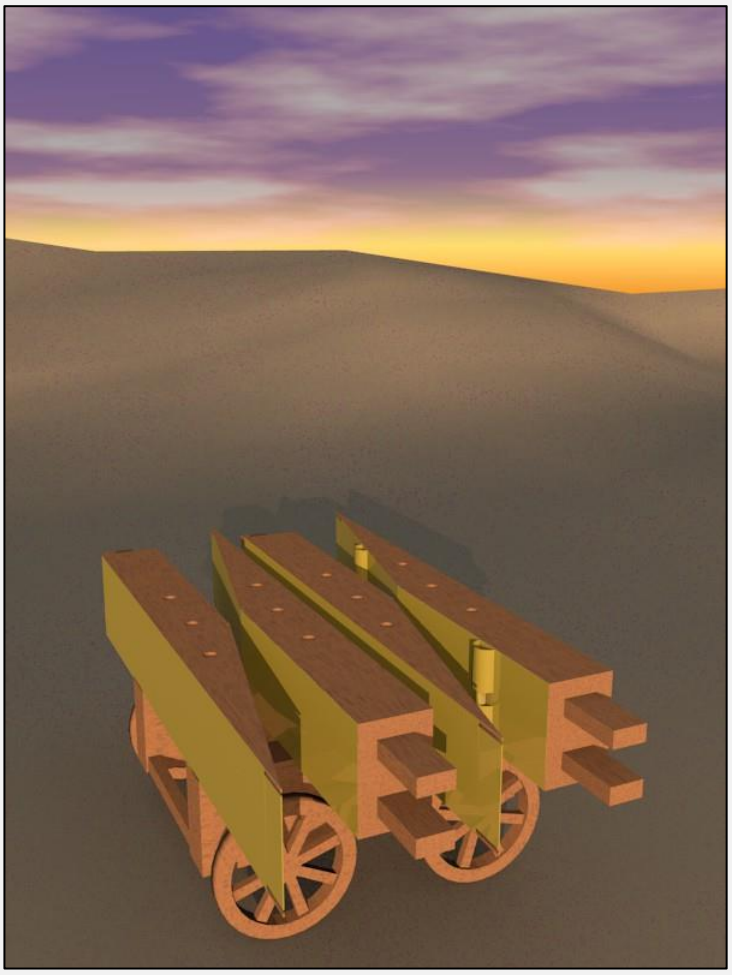
1. It is improper to place the bottom of one beam next to the top of another beam, similar to the idea that we always go up in holiness, but not down. Therefore, they wouldn't have done it that way.

2. They laid the beams down from their standing positions directly onto the wagons and it would have been more difficult to flip the ends of the beams around than to just stack them up one more layer.





3. Perhaps, according to Rebbe Yehuda, the beams only began to narrow from the midpoint of its height.

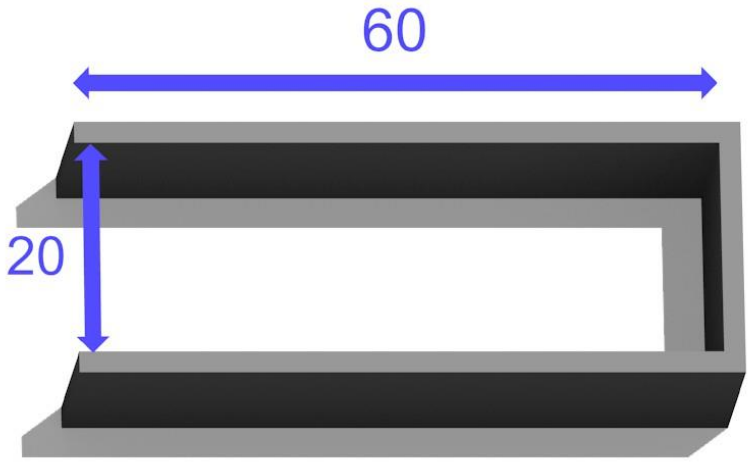


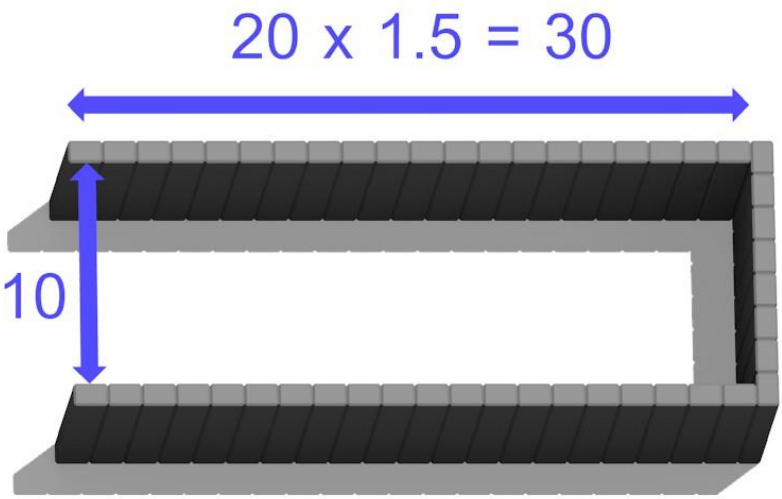
If this is so then it would have been pointless to flip the beams around, as they still would not have been able to fit any more than four beams along the length of the wagon.

Notes

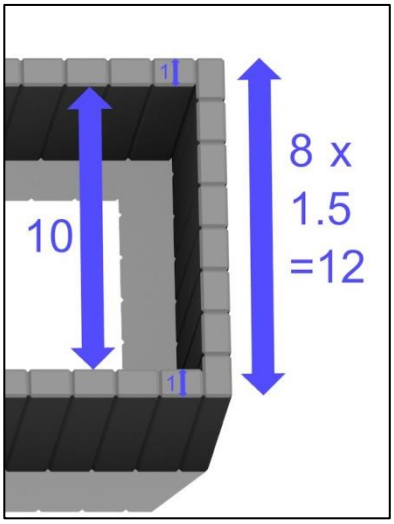
1. Exodus 26:16

2. There are a number of opinions as to the source of this. One approach is that since the area in the Third Temple corresponding to the area within the Mishkan walls had a length-to-width ratio of 3 to 1 (60 amos by 20 amos) (Ezekiel 41:2,4), it is logical to assume that the area within the Mishkan had the same ratio (Tosafos, Masechta Shabbos, page 98b).





3. Thus, since the length of the Mishkan was 30 amos (20 beams (Exodus 26:18) that were each 1.5 amos wide) it follows that the width of the Mishkan was 10 amos.



Now, since the back wall was 12 amos across (8 beams (Exodus 26:25)) it must be that the North and South walls were each 1 amah thick, giving us the thickness of the beams. (Rabbeinu Prachya, on Masechta Shabbos, page 96b, paragraph "And we say…")

3a & 3b.

The debate between Rebbe Nechemia and Rebbe Yehuda is based on the different ways to interpret the verse in Exodus 26:24. The plain reading of the verse is *"They shall be even at the bottom, and together shall they match at its top…"*

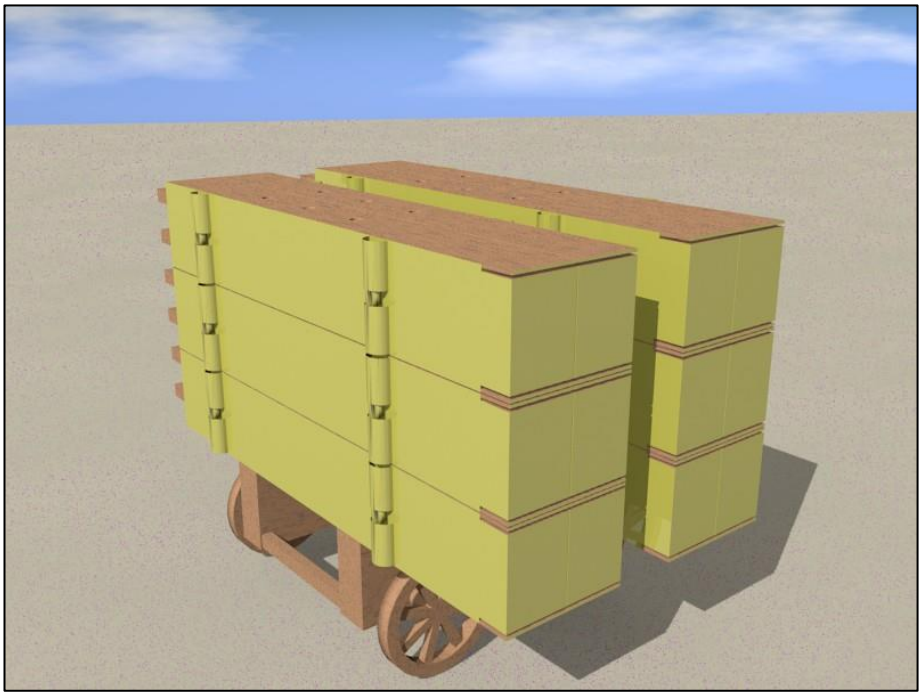
Rebbe Nechemia maintains that the word *"together"* implies that they should be at the top the same way that they are on the bottom, 1 amah thick.

Rebbe Yehuda focuses on the word "*match*", which in Hebrew (*tamim,* תמים) is the same root word as that which appears in Joshua 3:16 (*tamu,* תמו), where in that context it means *"cut off"*. Thus, he would read our verse as *"…and together they shall be 'cut off' at its top…"*

4. Masechta Shabbos 99a.

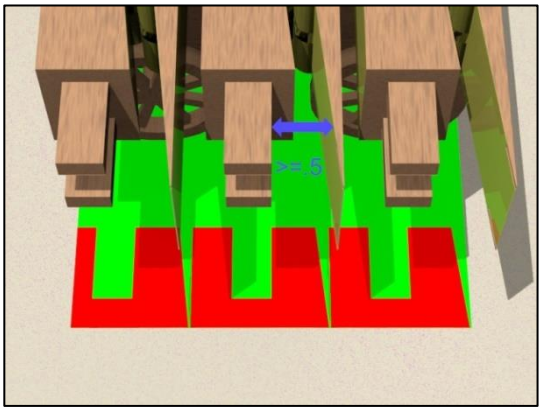
5. Masechta Shabbos 97a. The concept of *"extension"* means that we consider gaps that are less than a certain size to be legally negligible. (This is a logical necessity in every legal system. Basically, it is impossible to construct dimensions in the physical world conforming to an absolute, Euclidean specification. Even with modern technology, we cannot flatten the atom nor completely close the gaps between them. However, the Gemara finds this logic insufficient to totally explain this law, see there.)

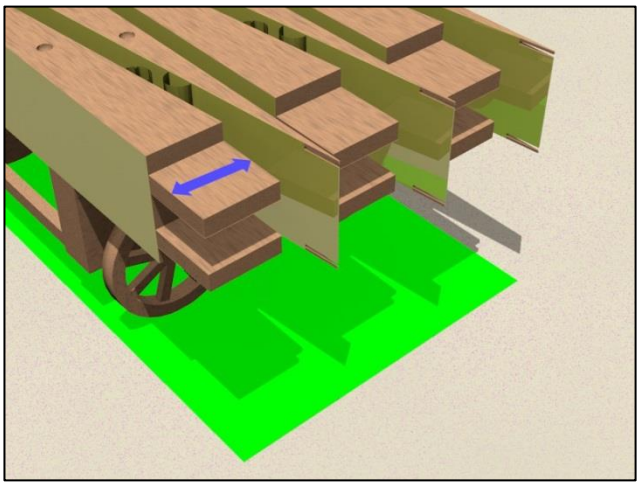
6. This was done for stability since they needed to stack three layers of beams on the wagon, for a total of 12 beams on each wagon, since there were 48 beams (Exodus 26:18-23) for 4 wagons (Number 7:8, 4:29-31).



7. Under the heading of "…הניחא", "It is well…"

8. Even though there is a small area at the edge of the beams that are too wide for *"extension"*. It could be that this area is definitely not what Rav was referring to since it is far from the wagon, and thus cannot be referred to as its "sides".





Another explanation might be that Tosafos holds like those opinions (Riva, Tur, Abarbanel, and others) that say that the tenons of the beams (the feet) went up to the edge of the beams' front and back faces, eliminating any area as large as 1/2 of an amah. *(The picture is the opinion of the Riva's explanation of Rashi).*

**Known error: The picture of the beams tapering off appears much narrower than it actually should be. This doesn't matter for the purposes of this presentation.*

All pictures were made with Sketchup and Blender.

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2015

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